without risking human pilots. This function was critical for improving the accuracy of anti-aircraft artillery during World War I and the interwar period. A notable example is the de Havilland DH.82B “Queen Bee” (1935), devel-oped by the British Royal Navy and Royal Air Force. It was a radio-con-trolled aircraft used to train gunners in tracking and targeting moving objects. The “Queen Bee” became so well known that the term “*drone*” was later derived from its name, symbolizing pilotless aerial vehicles.

2.5.2 Reconnaissance and surveillance missions

Although early drones lacked advanced cameras or sensors, military researchers began experimenting with remote reconnaissance using lightweight cameras or observation payloads. These efforts aimed to gather intelligence without endangering pilots. Early prototypes like the Aerial Target (1916) in the United Kingdom and the Kettering Bug (1918) in the United States were designed for short-range missions to observe enemy movements or deliver basic aerial imagery. While their accuracy and endurance were limited, these projects introduced the concept of unmanned intelligence gathering, which would later evolve into modern aerial surveillance.

2.5.3 Aerial bombing experiments

Another early military application of drones was in aerial bombing and guided munitions. The Kettering Bug, developed by Charles Kettering during World War I, was one of the earliest examples of an unmanned aerial torpedo. It was pre-programed using a mechanical timer to release its payload after flying a predetermined distance. Although it lacked real-time control and accuracy, the project demonstrated the potential of unmanned